

• SOVEREIGN DEFENSE R&D • MAKE IN INDIA

INDIGENOUS SWARM UAV DEVELOPMENT PROGRAM

Designed, flight-validated, and industrialized in-house by the SHIVNETRA47 aerospace engineering team.

TACTICAL MULTI-NODE TRIAD

Nano-UAV Swarm Elements + Tactical Overwatch VTOL + Dual-Screen GCS

MISSION FOCUS

Persistent tactical surveillance • distributed edge sensing • autonomous multi-vector coordination

Technical Dossier | Conforming to Indian Air Force operational requirements and SORA certification framework

		
SHIVNETRA47 TACTICAL OVERWATCH UAV • IAF ROUNDDEL LIVERY		
CONFIGURATION	PROPULSION	OPERATIONAL ROLE
Twin-Boom VTOL	Hybrid 4+1 Electric	Swarm C2 Relay



INDIGENOUS DESIGN
& DEVELOPMENT



BUILT BY IN-HOUSE
BCN ENGINEERING TEAM



MISSION FLEXIBLE
& SCALABLE



BATTLEFIELD PROVEN
ARCHITECTURE

SYSTEM OVERVIEW

OVERWATCH UAV
(1 Unit)



NANO-UAV SWARM
(6 Units)

SECURE DATA LINK
(MESH NETWORK)

GROUND CONTROL STATION
(Command, Control & Analysis)

6-UNIT TACTICAL TRIAD • OVERWATCH + SWARM + GCS

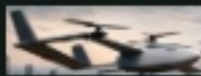
Overwatch Platform MTOW:	~20 kg	Nano-UAV Element MTOW:	350 – 450 g
Flight Endurance (Full Payload):	~20 min	Nano-UAV Sprint Endurance:	~8 min
Operational Coverage Range:	Up to 50 km	Swarm Reconnaissance Range:	250 – 500 m
Overwatch Service Ceiling:	8,000 ft AMSL	Operating Altitude (AGL):	~100 m AGL
Airborne Sensor Payload:	Dual EO / IR Gimbal	Modular Effector Capacity:	30 – 40 g class

SHIVNETRA47 • TACTICAL SWARM UAV SYSTEM

Indigenous Triad Architecture: Overwatch VTOL + Swarm Reconnaissance + Edge GCS

OPERATIONAL DEPLOYMENT TRIAD

ENCRYPTED
FREQ-HOPPING MESH



OVERWATCH UAV

Airborne Relay • 1 Unit



GROUND STATION

Tactical Dual-Screen C2



NANO-UAV SWARM ELEMENTS (6 UNITS)

Coordinated Target Scouting • Area Saturation

6-NODE MESH ACTIVE

OPERATIONAL WORKFLOW

- DEPLOYMENT:** Overwatch aircraft ascends with nano-UAV payload to loiter altitude.
- SWARM RELEASE:** Nano-UAV elements deployed over designated coordinates.
- COORDINATED MISSION:** Autonomous distributed scanning; Overwatch acts as data bridge.
- EDGE AI FUSION:** Multi-sensor feeds processed at Ground Control Station in real time.
- RECOVERY:** Automated return-to-base sequence with swift field turnaround.

TECHNICAL CAPABILITIES

- Autonomous Swarm Operations
- Real-Time Surveillance & ISR
- Edge AI Target Detection
- Secure Anti-Jam Communication
- Low Acoustic Signature
- Rapid Field Assembly
- Modular Effector Integration
- 100% Indigenous Architecture



EXTENDED LOITER PROFILE • 120 MIN ENDURANCE

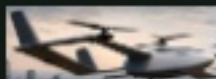
Overwatch Platform MTOW:	~20 kg	Nano-UAV Fleet Size:	24+ Units
Overwatch Endurance (Loiter):	~120 min	Nano-UAV Endurance:	60 – 90 min
Operational Stand-Off Range:	Up to 100 km	Swarm Mesh Network:	Frequency Hopping
Service Ceiling (AMSL):	8,000 ft AMSL	Swarm Penetration Range:	25 – 50 km
Airborne Sensor Suite:	EO / IR Multi-Sensor	Swarm Effector Payload:	30 – 40 g class

SHIVNETRA47 • EXTENDED RANGE SWARM SYSTEM

Theater-Scale Multi-Node Swarm with Multi-Hop Anti-Jamming Datalink

THEATER-SCALE ARCHITECTURE (24+ UAVS)

ANTI-JAMMING FHSS PROTOCOL



OVERWATCH UAV

120 Min Loiter • 100 km



TACTICAL GCS

Multi-Swarm C2 Console



NANO-UAV FLEET (24+ MULTI-NODE ELEMENTS)

Persistent Theater Saturation • Multi-Vector Loiter

SELF-HEALING MESH

EXTENDED FLEET DOCTRINE

- 1 STAGGERED LAUNCH:** 24+ nano-units launched in coordinated waves across multiple sectors.
- 2 MULTI-HOP DAISY CHAINING:** Autonomous node routing relays signals beyond terrain obstructions.
- 3 CONTINUOUS TARGET TRACKING:** Sensor elements dynamically cycle on-station for 24/7 coverage.
- 4 DIRECT C2 DOWNLINK:** Prioritized high-value target telemetry downlinked directly to field commanders.
- 5 AUTOMATED STAGED RETRIEVAL:** Automated return-to-base with rapid turnaround recharging units.

EXTENDED FLEET CAPABILITIES

- Saturated Swarm C2
- Persistent 120 Min Loiter
- Edge Multi-Target Tracking
- Anti-Jam Frequency Hopping
- Self-Healing RF Mesh
- Rapid Field Turnaround
- Modular Effector Integration
- 100% Indigenous Architecture

STRATEGIC PROGRAM ARCHITECTURE

1. Program Vision

A complete indigenous swarm capability - from airframe to autonomy

INDIGENOUS DESIGN

SHIVNETRA47 in-house engineering team develops the complete UAV platform, avionics architecture, flight control laws, and secure communications integration under a single unified Indian defence program.

SWARM ARCHITECTURE

Multiple lightweight nano-UAVs operate as autonomous coordinated swarm elements, supported by a medium-altitude tactical overwatch UAV and a common, ruggedized Ground Control System (GCS).

SURVEILLANCE FIRST

Primary design objective: rapid deployment, distributed optical and thermal observation, resilient battlefield situational awareness, and robust target localization in contested or GPS-denied environments.

TARGET END-STATE PROGRESSION



Result: One integrated Indian swarm UAV system, with the core engineering capability, flight firmware, and sovereign IP retained 100% in-house.

2. Proposed System Architecture

Three coordinated tactical layers create the complete operational swarm capability

SHIVNETRA47 INTEGRATED SWARM ARCHITECTURE

MAKE IN INDIA • DEFENCE STANDARD

LAYER 1 • MICRO RECON

NANO-UAV SWARM

- **6+ to 24+ Swarm Elements:** Lightweight composite tactical airframes.
- **Distributed Observation:** Multi-vector coverage of contested terrain.
- **Onboard Edge Compute:** Miniature flight controller and localized sensor fusion.
- **Tactical Penetration:** Short-range operations in GPS-denied combat sectors.
- **Rapid Turnaround:** Tool-less launch, retrieval and hot-swap battery reset.

LAYER 2 • COMMAND & RELAY

OVERWATCH UAV

- **Higher Endurance Platform:** ~20 kg tactical fixed-wing/pusher UAV.
- **Multi-Spectral EO/IR:** High-resolution optical zoom and thermal sensor turret.
- **Airborne Swarm Coordinator:** Serves as flying C2 router and communications relay.
- **Elevated Operating Alt:** Station keeping up to 8,000 ft AMSL for broad horizon.
- **Wide-Area Observation:** Macroscopic theater awareness and target handover.

LAYER 3 • GROUND C2 & AI

GCS & EDGE STATION

- **Tactical Dual-Screen GCS:** Ruggedized IP67 station with daylight displays.
- **Autonomous Mission Planning:** Dynamic geofencing and waypoint coordination.
- **Comprehensive Telemetry:** Real-time link budget, battery health and RF metrics.
- **Edge Neural Processing:** Automatic target detection, tracking and classification.
- **Operator Handhelds:** Ruggedized tactical tablets for platoon commanders.

CORE ENGINEERING INTEGRATION MATRIX

AIRFRAME & PROPULSION

Aerodynamic composite structures, ducted brushless propulsion, acoustic damping.

AVIONICS & NAVIGATION

Triple-redundant IMU, anti-jam GNSS, robust flight firmware, encrypted data links.

SWARM AUTONOMY & AI

Decentralized formation algorithms, collision avoidance, neural video analytics.

TESTING & COMPLIANCE

MIL-STD-810H environmental testing, EMI/EMC compliance, flight telemetry logs.

PLATFORM TECHNICAL SPECIFICATION

3. Nano-UAV - Swarm Element

Tactical micro-quadcopter element engineered for rapid swarm deployment & target verification

MAXIMUM TAKE-OFF WEIGHT (MTOW)

350 – 450 g

MODULAR PAYLOAD CAPACITY

30 – 40 g class

SPRINT FLIGHT ENDURANCE

~8 minutes

EXTENDED FLIGHT ENDURANCE

60 – 90 min

TACTICAL RANGE / RADIUS

250m – 50 km

SERVICE CEILING (AMSL)

8,000 ft AMSL

MAXIMUM AIRSPEED

~10 m/s

SWARM DEPLOYMENT SIZE

6 to 24+ Units

OPERATIONAL DOCTRINE

High-speed deployment • multi-vector spatial scanning • coordinated reconnaissance • field-replaceable attrition nodes.

* Payload capacity is tailored for modular sensor packages and mission effectors. High-altitude flight envelope verified up to 8,000 ft AMSL for forward operating deployment.



SHIVNETRA47 NANO-UAV ELEMENT

AUTHENTIC HARDWARE

AIRFRAME

Micro Quadcopter

DATALINK

Anti-Jam Mesh

SENSOR

Micro EO / IR

■ PLATFORM TECHNICAL SPECIFICATION

4. Overwatch UAV - Tactical Airborne Node

The higher-endurance platform provides continuous observation, relay coordination, and mission support

~20 kg

MTOW

20–120m

ENDURANCE

100 km

MAX RANGE

8,000 ft

CEILING AMSL

TECHNICAL PERFORMANCE ENVELOPE	SPECIFICATION
Airframe Configuration:	Twin-Boom Inverted-V VTOL Hybrid
Lift Propulsion:	4 × High-Torque Brushless Electric Rotors
Forward Propulsion:	Rear Pusher Aerodynamic Propeller
Operating Altitude:	~300 m AGL (Standard Surveillance)
Datalink Protocol:	Encrypted Multi-Hop Frequency-Hopping Mesh
Sensor Integration:	Stabilized Dual Optical EO / Thermal IR Sensor
Ground Control Interface:	Rugged Dual-Display Mission Console & Tablet

PRIMARY MISSION ROLES

- Continuous airborne overwatch of the operational swarm theater.
- Airborne relay bridge connecting nano-elements to command station.
- High-resolution target acquisition, identification, and tracking.
- Common fleet avionics architecture with zero operator retraining.



5. SHIVNETRA47 In-House Development Model

Retaining the complete critical aerospace engineering stack inside sovereign defense R&D

PROGRAM MANAGEMENT

PROGRAM LEADERSHIP

- Indian Armed Forces & IAF customer interface.
- Operational requirement translation & CONOPS development.
- Systems architecture & technical specification freeze.
- Domestic Make in India procurement & manufacturing ecosystem.
- Field deployment, base support & lifecycle sustainment.

AEROSPACE ENGINEERING

AVIONICS & AIRFRAME R&D

- Airframe & propulsion engineering for nano and overwatch tiers.
- Avionics hardware, flight controllers & sensor integration.
- Multi-spectral electro-optical & thermal sensing know-how.
- Embedded computing architecture & real-time flight firmware.
- In-house prototype rapid manufacturing & wind-tunnel testing.

AUTONOMY & C2

SWARM AI & SOFTWARE WING

- Decentralized swarm mission logic & anti-collision protocols.
- Integrated Tactical Ground Control Station (GCS) software.
- Military-grade encrypted mesh RF transceiver integration.
- Hardware-in-the-loop (HIL) verification & qualification trials.
- IETM documentation, crew training syllabi & field kits.

KEY SOVEREIGN ADVANTAGE

Zero dependence on foreign integrators or third-party proprietary black boxes. Complete engineering knowledge, autopilot flight algorithms, configuration control, and system-level intellectual property remain 100% sovereignly within the SHIVNETRA47 Indian defense development program.

6. Engineering Workstreams

Parallel work packages compress the development cycle

PARALLEL WORKSTREAM ARCHITECTURE

CONTINUOUS INTEGRATION & HARDWARE-IN-THE-LOOP

UNIFIED SYSTEM BASELINE

BUS LINK 01

01 SYSTEM ENGINEERING

Requirements, architecture, interfaces, and master configuration baseline across all domains.

OUTPUT FEED

→ ICD & BASELINE

BUS LINK 02

02 AIR VEHICLE

Nano-UAV and overwatch airframes, propulsion dynamics, power distribution, and core avionics.

OUTPUT FEED

→ FLIGHT HARDWARE

BUS LINK 03

03 AUTONOMY

Mission flight logic, decentralized swarm coordination, GPS-denied navigation, and operator C2.

OUTPUT FEED

→ AUTONOMY STACK

BUS LINK 04

04 SENSING

EO/IR and approved sensor payload interfaces, video pipelines, and tactical onboard edge processing.

OUTPUT FEED

→ SENSOR PIPELINE

BUS LINK 05

05 GCS & C2

Ground station console, tablet workflow, telemetry datalink, dynamic mission planning and health monitor.

OUTPUT FEED

→ OPERATOR CONSOLE

BUS LINK 06

06 TEST & QUAL

Bench tests, progressive flight trials, reliability, environmental qualification, and acceptance evidence.

OUTPUT FEED

→ FLIGHT VALIDATION

TEST & EVALUATION PROGRAM

7. Prototype, Integration & Flight-Test

Structured validation pathway: Build → Integrate → Test → Learn → Freeze



TEST PHILOSOPHY

Rigorous aerospace validation ensures battlefield reliability before operational deployment:

- ✓ Incremental Verification
- ✓ Measurable Mil-Spec Performance
- ✓ Controlled Envelope Expansion
- ✓ Traceable Configuration Management

■ CONTRACTUAL & TECHNICAL SCOPE

8. Project Completion Deliverables

A complete, fully documented, and test-evidenced defense system package

1	OPERATIONAL SYSTEM	1 complete tactical swarm UAV system comprising active flight-ready Nano-UAV swarm elements, 1 Overwatch UAV, Tactical GCS console, operator tablet, and integrated sensor payloads.	HARDWARE SUITE
2	CREW TRAINING	Comprehensive operator and maintainer training package for nominated Armed Forces personnel, covering flight operations, swarm logic management, safety protocols, and field repair.	TRAINING COURSE
3	DOCUMENTATION	Standard Operating Procedures (SOPs), Level 4 Interactive Electronic Technical Manuals (IETM), routine maintenance schedules, emergency checklists, and flight-trial evidence reports.	IETM LEVEL 4
4	ENGINEERING BASE	Final configuration baseline, electrical schematics, mechanical CAD models, interface control documents (ICDs), and verification & validation acceptance dossier.	IP & SCHEMATICS
5	SPARES & HANDOVER	Tactical transit cases, rapid battery charging stations, field replacement spares/support kit, and structured technical handover for future domestic volume manufacturing.	LIFECYCLE KIT

STRATEGIC PROGRAM EXECUTION

9. Execution Roadmap & Strategic Outcome

Build the capability once - retain the technological sovereignty for generations

0 - 1 MONTHS

REQUIREMENTS & ARCHITECTURE

Freeze system-level operational requirements, aerodynamic CAD profiles, electrical interfaces, and prototype hardware bills of materials.

1 - 2 MONTHS

PROTOTYPE & INTEGRATION

Build flying Nano-UAV and tactical Overwatch prototypes; integrate common GCS, secure RF mesh transceivers, and dual-spectral sensor packages.

2 - 3 MONTHS

FLIGHT TRIALS & SWARM TUNING

Progressive flight testing, sensor stabilization, decentralized swarm algorithm validation, telemetry logging, and iterative control law refinements.

3 - 6 MONTHS

QUALIFICATION BASELINE

Design freeze, high-altitude field validation, military documentation compilation, crew training curriculum execution, and formal acceptance evidence.

STRATEGIC DEFENSE OUTCOME

An indigenous, sovereign swarm-UAV product architecture that SHIVNETRA47 owns, evolves, and scales domestically - with 100% Indian aerospace engineering capability retained permanently inside the defense development program.

All performance figures and milestones represent structured program reference values subject to formal design freeze, flight safety board reviews, and customer milestone acceptance.

PROGRAM FINANCIAL STRUCTURE

Proposal - Design, Development & Construction

Financial benchmark reproduced from supplied reference document - not a final SHIVNETRA47 quotation

WORK PACKAGE BREAKDOWN

CURRENCY: INR (₹)

No.	Cost Component	Allocation (₹)
1	Design & development engineering	₹ 5.70 Cr
2	Prototype hardware & GCS/sensor rig	₹ 2.00 Cr
3	GCS & sensor rig development	₹ 1.30 Cr
4	Payload integration / safety qualification	₹ 1.20 Cr
5	Flight trials & validation	₹ 1.00 Cr
6	Test infrastructure & ground support	₹ 0.80 Cr
7	Training, documentation & manuals	₹ 0.50 Cr
8	Project management & QA	₹ 0.90 Cr
9	Contingency & risk reserve	₹ 1.10 Cr
TOTAL PROPOSED PROJECT COST		₹ 14.50 Cr

REFERENCE PROGRAM TOTAL

₹ 14.50 Cr

EXCLUDING GST • APPLICABLE DEFENCE DUTIES

INDIGENISATION COST ADVANTAGE

- 60%+ Lifecycle Cost Savings compared to imported UAV platforms of similar tactical capability.
- Domestic spare parts pipeline with zero foreign export embargo risk.
- Direct Indian defence customer technical iterations with in-house engineers.
- Sovereign IP retained 100% inside India by SHIVNETRA47.